

Functions & Graphs



Function transformations

- 1 Let $f(x) = x^2$ and $g(x) = -3f(x + 2) - 4 = -3(x + 2)^2 - 4$. Find $g(0)$, and describe the transformation from f to g in words.
 - 2 Using the same $g(x) = -3(x + 2)^2 - 4$, find $g(-2)$, and explain why this is the vertex of g 's graph.
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Even and odd functions

- 3 Classify $f(x) = x^4 + 2x^2 - 5$ as even, odd, or neither.
 - 4 Classify $h(x) = 2x^5 - 3x$ as even, odd, or neither.
 - 5 Explain why $k(x) = x^2 + x$ is neither even nor odd.
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Domain restrictions

- 6 Find the domain of $f(x) = \sqrt{2x + 6}$.
 - 7 Find the domain of $g(x) = \frac{1}{x^2 - 25}$.
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Piecewise functions

- 8 Let $p(x) = 2x + 3$ for $x < 1$, and $p(x) = x^2 - 1$ for $x \geq 1$. Find $p(-2)$ and $p(3)$.